CSA0716 - COMPUTER NETWORKS

6G Communication Architecture Compared with the OSI Model

Presented By

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ABSTRACT

- **6G** provides ultra-fast, low-latency, and intelligent wireless communication.
- The **OSI model** explains network communication using seven layers.
- This project compares **6G architecture** with the **OSI model**.
- It highlights technologies like **AI, THz, Edge Computing, and IoE**.
- The comparison shows how **6G improves future network performance and efficiency**.

INTRODUCTION

- **6G** is the next generation of wireless communication technology.
- The **OSI model** explains communication through seven network layers.
- This project compares **6G architecture** with the **OSI model**. 6G uses **AI, THz, Edge Computing, and IoE** technologies.
- The comparison shows the future of **smart and efficient communication networks**.

OBJECTIVE

- To understand the **6G communication architecture.**
- To study the **OSI seven-layer model.**
- To compare **6G architecture** with the **OSI model.**
- To identify the role of **advanced 6G technologies.**
- To analyze the benefits of **6G in future communication networks.**

METHODOLOGY

- Study the **6G communication architecture**.
- Analyze the **OSI seven-layer model**.
- Compare the functions of **6G and OSI layers**.
- Identify key **6G technologies and features**.
- Evaluate the comparison and summarize the results.

OSI Layer	6G Communication Architecture (What it Does)	Key 6G Technologies (Examples)
7 Application User needs and services	• Provides end-user services and applications	• AR/VR, Holograms • Smart Cities, Healthcare • Autonomous Vehicles
6 Presentation Formats data for use	• Formats, encrypts and compresses data for secure transmission	• Encryption • Compression • Data Encoding
5 Session Manages sessions and security	• Establishes, maintains and terminates sessions • Ensures security and authentication	• Authentication • Session Management • Security Services
4 Transport Ensures reliable end-to-end delivery	• Ensures reliable, low-latency and quality communication between endpoints	• QoS, Ultra-Low Latency • Reliable Transmission • Flow Control
3 Network Routes data between networks	• Routes data using intelligent networking and different segments	• AI Routing, IPv6 • Network Slicing, Edge Computing • SDN/NFV
2 Data Link Ensures data link between devices	• Manages access to the medium and detects/corrects errors	• MAC Protocol • Error Control • Resource Allocation
1 Physical Transmits raw bits over medium	• Transmits raw data over physical channels using advanced radio technologies	• THz Spectrum, Massive MIMO • Beamforming, RIS • Ultra High-Speed Connectivity

★ 6G: Connecting Everything Intelligently for a Sustainable and Intelligent Future ★

MODULE 1

Module 1: Study of the OSI Model and Study of 6G Communication

OSI Model	6G Communication
7-layer network communication model.	Next-generation wireless technology.
Standardizes data transmission.	Provides ultra-fast, low-latency communication.
Ensures reliable networking.	Uses AI, THz, and Edge Computing.
Used in all computer networks.	Supports IoT, Smart Cities, XR, and autonomous vehicles.

MODULE 2

Model Development

Module 2: Comparison of 6G Architecture with the OSI Model

- Compare the **6G communication architecture** with the **OSI model.**
- Identify the role of each OSI layer in 6G communication.
- Analyze the similarities and differences between both architectures.
- Evaluate the advantages and future scope of 6G communication networks.

ADVANTAGES

- **Ultra-high data transmission speed** for faster communication.
- **Very low latency** for real-time applications. **AI-based intelligent network management** improves efficiency
- **Enhanced security and reliability** for safe data communication.
- **Supports future technologies** such as smart cities, IoE, autonomous vehicles, and holographic communication.

APPLICATION

- **Smart Cities** – Supports intelligent traffic and public services.
- **Autonomous Vehicles** – Enables real-time communication for self-driving cars.
- **Healthcare** – Supports remote surgery and telemedicine.
- **Internet of Everything (IoE)** – Connects smart devices, sensors, and machines.
- **Industrial Automation** – Improves smart factories and robotics with high-speed connectivity.

FUTURE ENHANCEMENT

Improve AI-based network automation for smarter communication.

Increase network speed and capacity using advanced 6G technologies.

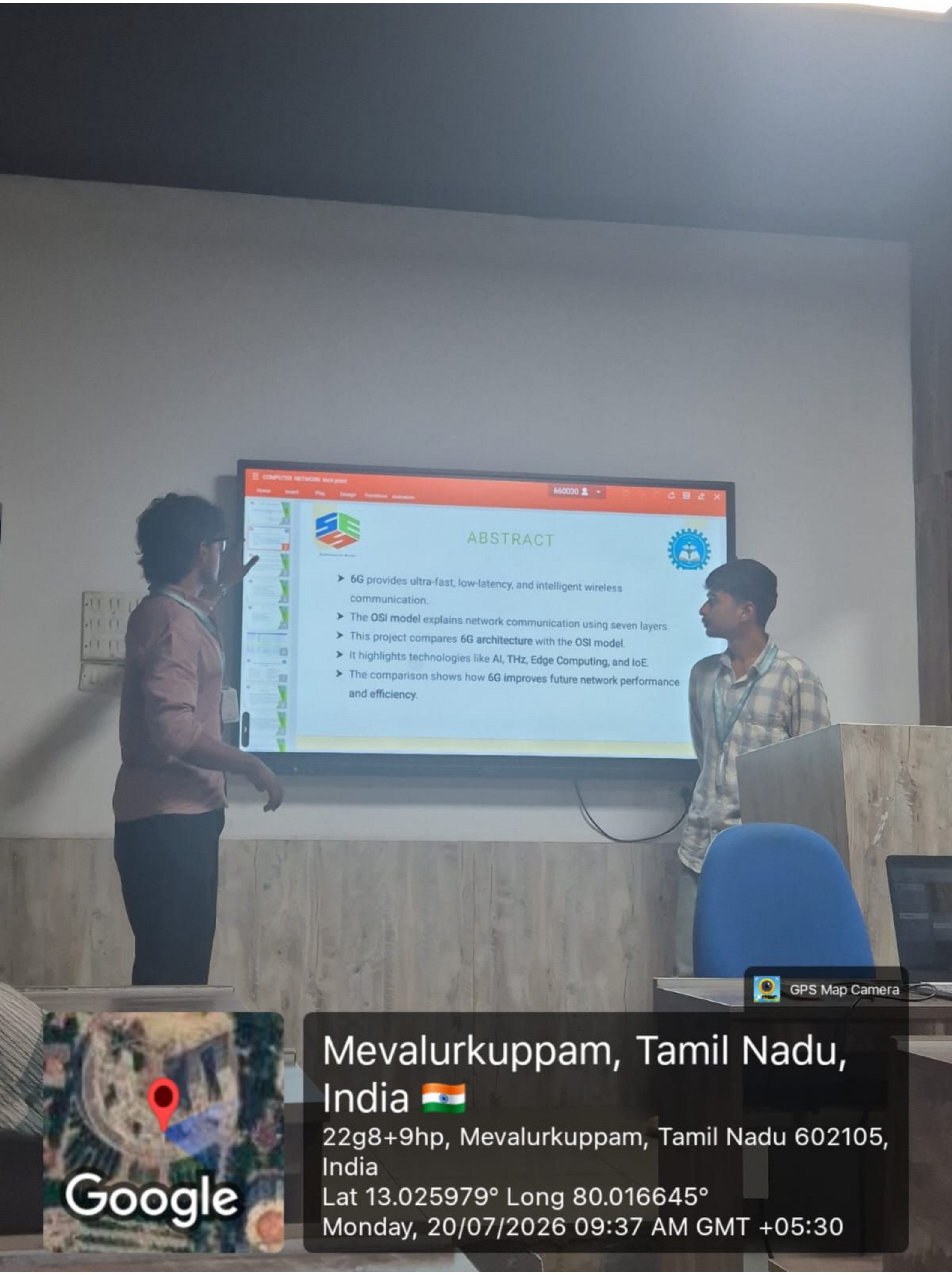
Enhance security and privacy with next-generation encryption.

Expand support for IoE and smart devices across various industries.

Enable advanced applications such as holographic communication and digital twins.

CONCLUSION

- **6G** is the future of wireless communication with high speed and low latency.
- The **OSI model** provides a standard framework for network communication.
- Comparing **6G and the OSI model** helps understand modern networking concepts.
- **AI, THz communication, and Edge Computing** make 6G more intelligent and efficient.
- **6G will enable faster, smarter, and more reliable communication networks** in the future.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The OSI model explains network communication using seven layers.
This project compares 6G architecture with the OSI model.
It highlights technologies like AI, THz, Edge Computing, and IoE.
The comparison shows how 6G improves future network performance and efficiency.
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Thank You